A Diff-Attention Aware State-Space Fusion Model for Remote Sensing Classification

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***Abstract*—Multispectral (MS) and panchromatic (PAN) images describe the same land surface, so these images not only have their own advantages but also share a significant amount of redundant information. In order to separate similar information and each modality's unique advantages, thereby reducing feature redundancy at the fusion stage, this article introduces a diff-attention-aware state-space fusion model (DASF-Model) for multimodal remote sensing (RS) image classification. Based on the selective state-space model (SSM), a cross-modal diff-attention module (CDAM) is designed to extract and separate the common features and their respective dominant features of MS and PAN images. Specifically, space preserving Visual Mamba (SPVM) retains image spatial features and captures local features by appropriately optimizing Visual Mamba's input. Considering that features in the fusion stage will have large semantic differences after feature separation and the traditional mean fusion method fails to effectively integrate these features with significant discrepancies, an attention-aware linear fusion module (ALFM) is proposed. It performs pixelwise linear fusion by calculating influence coefficients. This mechanism can fuse features with large semantic differences while keeping the feature size unchanged. Empirical evaluations indicate that the presented method achieves better results than alternative approaches. The relevant code can be found at: https://github.com/AVKSKVL/DAS-F-Model**

***Index Terms*—Feature fusion, feature separation, fusion classification, Mamba, multimodal.**

I. Introduction

In RECENT years, improving remote sensing (RS) image classification accuracy through multisource fusion has become a key research focus. With the rapid advancement of satellite technology, the demand for high-quality RS images that combine both high spatial resolution and multispectral (MS) capabilities has increased significantly [1], [2]. Modern satellites can simultaneously capture panchromatic (PAN) images with very high spatial resolution and MS images, which provide rich spectral information [3], [4]. However, while MS images offer richer spectral information [5], their spatial resolution is typically lower than that of PAN images. The combination of these complementary features enables effective fusion and classification for various practical applications, such as deforestation monitoring [6], urban planning [7], precision agriculture [8], land-cover classification [9], [10], and mineral exploration [11], [12].

In RS, multisource classification algorithms fall into two main categories: traditional methods and deep learning-based algorithms [13], [14], [15], [16], [17].

One traditional approach is to apply PAN sharpening [18], [19], [20] techniques to enhance MS images before classification. These algorithms require appropriate PAN-sharpening techniques, usually, such as component substitution (CS) [21] and multiresolution analysis (MRA) [22]. Another traditional approach is feature fusion followed by classification, where features are extracted separately from PAN and MS images. However, these conventional methods often rely on handcrafted features and manually designed fusion rules, which may not effectively capture the complex relationships between multimodal data.

With the rapid development of deep learning theory and hardware capabilities, significant progress has been achieved in deep learning-based multimodal fusion classification algorithms. Among these, convolutional neural network (CNN) and Transformer architectures have emerged as the predominant research directions.

Based on the CNN method, Ma et al. [23] introduced the interactive spatial–spectral perception (ISSP) network. This network addressed the challenge of optimizing complementary information between PAN and MS images by enhancing spatial location interaction and integrating multiscale context through pixel-guided spatial enhancement. Lu et al. [24] proposed coupled adversarial learning for classification, which learned more discriminative fusion features by employing an optimally designed joint loss function that included both unsupervised adversarial and supervised classification losses. However, their ability to capture global information was somewhat limited, restricting the model's capacity to analyze complex scenes. Based on the Transformer architecture, Roy et al. [25] introduced a novel multimodal fusion Transformer (MFT) network, which employs multihead cross-patch attention to learn distinctive representations in simplified and hierarchical feature spaces. The S^2ENet [26] incorporated a spatial enhancement module for LiDAR image and a

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spectral enhancement module for HS image, enhancing their respective spatial and spectral features. However, although Transformer-based fusion classification models demonstrate significant potential in feature extraction and fusion, their high computational cost and optimization difficulty remain critical challenges.

The state-space model (SSM)-based Mamba architecture [27] achieves efficient global sequence modeling with linear time complexity. Among its key designs, the 1-D convolutional design compensates for the SSM's deficiency in local feature modeling. As a result, Mamba [27], which combines the advantages of both CNN and Transformer, has quickly gained attention in the field of natural language processing. Visual Mamba [28] extends the Mamba architecture to the visual domain by partitioning images into patch-based sequences and replacing 1-D convolution with depthwise convolution (DWConv).

Although the above works have achieved considerable results, there are still many problems to be solved.

1) In the field from text to vision, simply extending 1-D convolutions to DWConvs will cause the convolved objects to be nonadjacent in physical space. This is not only unfavorable for the retention and extraction of spatial information but also introduces useless spatial noise.
2) There are a large number of similar redundant features in MS and PAN images. This results in important information that plays a key role in downstream tasks being submerged in these redundancies. Han et al. [29] proposed a style separation and mode unification network. It first separates styles using wavelet decomposition and adversarial learning, then unifies modalities through an encoder–decoder architecture. However, the shared characteristics between MS and PAN images may extend beyond mode features, and their respective advantageous features might not be limited to style characteristics alone. Consequently, this model still has room for improvement in feature redundancy reduction. Xu et al. [30] proposed a dual-stream Transformer based on diff-attention, which extracted the advantageous features and common essential features by using self-attention and co-attention, and reduced the essential redundancy in the advantageous features by diff-attention enhancement. However, constrained by the quadratic complexity of the Transformer, the model exhibits limited scalability for high-resolution image processing.
3) When fusing multimodal features, simple fusion operations are difficult to effectively fuse features with large differences. Even if gating-based methods [30] are applied, the huge difference between features is still difficult to effectively compensate.

In order to solve the above problems, this article proposes a diff-attention-aware state-space fusion model (DASF-Model) for RS classification. The main contributions are as follows.

1) For the effective preservation of spatial information, a space preserving Visual Mamba (SPVM) is introduced. By reasonably adjusting the input features of Visual Mamba, the DWConv can effectively capture the local features of images, which promotes the effective combination of the SSM and the traditional visual model.
2) To address the high redundancy in MS and PAN feature fusion, a cross-modal diff-attention module (CDAM) is proposed. This module separates the common and dominant features of MS and PAN images by exchanging a series of matrices in the SSM. This greatly reduces the redundancy of fusion features, which enables the DASF-Model to pay more attention to more important key information in subsequent classification and improves the performance of classification.
3) Given the difficulty in effectively fusing features with substantial differences, we propose an attention-aware linear fusion module (ALFM) in this article. The module performs pixelwise linear fusion by dynamically calculating influence coefficients, allowing one modality to guide the fusion process of the other at each spatial location. This mechanism preserves feature size while fusing features with large differences.

This article is structured as follows. Section II explains the relevant work, Section III introduces the DASF-Model algorithm comprehensively, Section IV demonstrates the DASF-Model's effectiveness via rigorous experiments, and Section V presents the conclusion and future research directions.

II. Related Works

This section will introduce two tasks related to our method in detail: the SSM and multimodal fusion classification.

A. State-Space Model

SSM is a mathematical model used to describe dynamic systems, which provides an efficient approach for sequence modeling through linear time-invariant systems. It maps a 1-D input sequence x(t) to a latent state h(t) before generating outputs y(t)

$$\begin{aligned} \mathrm{dh}(t)/\mathrm{dt} &= \mathrm{Ah}(t) + \mathrm{Bx}(t) \\ y(t) &= \mathrm{Ch}(t) + \mathrm{Dx}(t) \end{aligned} \tag{1}$$

where $A \in \mathbb{R}^{N\times N}$ is the state transition matrix, $D \in \mathbb{R}^{N}$ is the direct transfer matrix (which may not exist in some cases), and $B \in \mathbb{R}^{N\times 1}$ and $C \in \mathbb{R}^{1\times N}$ are the matrices responsible for transforming inputs into states and states into outputs, respectively.

For discrete sequences, the system is typically discretized using zero-order hold (ZOH) with step size Δ

$$\begin{aligned} h_k &= \bar{A}h_{k-1} + \bar{B}x_k \\ y_k &= Ch_k + Dx_k \end{aligned} \tag{2}$$

where $\bar{A} = \exp(\Delta A)$, $\bar{B} = (\Delta A)^{-1}(\bar{A} - I)\Delta B$, and Δ is a learnable parameter. In addition, the above formula can be mathematically equivalently converted into the following CNN form:

$$\begin{aligned} \bar{K} &\triangleq \left(C\bar{B}, C\overline{AB}, \ldots, C\bar{A}^{L-1}\bar{B}\right) \\ y &= x * \bar{K} \end{aligned} \tag{3}$$

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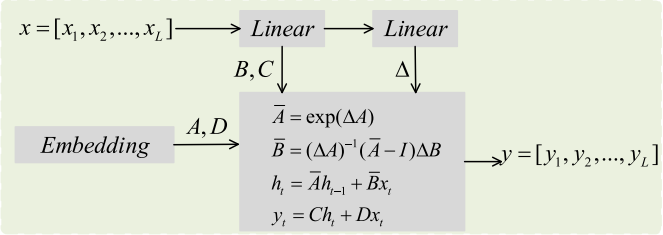


Fig. 1. Linear-time sequence modeling with selective state spaces (Mamba).

where L is the sequence length, * represents the convolution operation, and $\bar{K} \in \mathbb{R}^L$ is a structured convolution kernel. Although this model has linear computational complexity, its time-invariant parameter matrices (constant across all timesteps) inherently limit its capacity to capture sequence context.

On this basis, the S6 (Mamba [27]) model came into being. By introducing a selective mechanism (see Fig. 1), Mamba enables the model to dynamically adjust the state transition and observation process, so as to better adapt to different input characteristics and task requirements. Specifically, Mamba gets the parameter matrices B, C, and Δ based on the input $x \in \mathbb{R}^{b\times L\times D}$

$$\begin{aligned}(B, C, \Delta) &= \text{Linear}(x)\\ \Delta &= \text{Linear}(\Delta)\end{aligned} \tag{4}$$

where $\Delta \in \mathbb{R}^{b\times L\times D}$, $B \in \mathbb{R}^{b\times L\times N}$, and $C \in \mathbb{R}^{b\times L\times N}$. L means the sequence length and b is the batch size. Furthermore, this model has been applied in the field of multimodal fusion classification [31].

B. Multimodal Fusion Classification

Multimodal fusion classification is an emerging research field that focuses on integrating heterogeneous data from various sensors or sources. By leveraging the synergy and complementarity of these multimodal data, this approach significantly enhances both the accuracy and robustness of RS image classification.

For instance, Liu et al. [32] introduced the cross-resolution hidden layer feature fusion (CRHFF) method to better model objects using multimodal representations. By integrating multiscale features from MS and PAN images, their approach significantly boosts classification performance. Gao et al. [33] proposed a deep feature interaction network (DFINet), which is able to extract the autocorrelation and cross correlation of multisource features. At the same time, it is optimized by coordinating the consistency loss, discrimination loss, and classification loss. Wang et al. [34] proposed a nearest neighbor-based contrastive learning network (NNCNet), which makes full use of a large amount of unlabeled data to learn discriminative feature representations, effectively reducing the heterogeneous gap caused by the inconsistent distribution of multisource data.

However, none of the above methods takes into account that multimodal features may contain substantially similar redundant information, thereby failing to enable the model to focus on the critical information that determines the performance of classification tasks.

III. Methodology

To effectively separate the common essential features and their respective dominant features of MS and PAN images, while achieving effective fusion of these highly distinct features, this article proposes a DASF-Model (see Section III-A, Fig. 2) for multimodal RS image classification. The model consists of three modules: the SPVM [see Section III-B, Fig. 3(b) and 4], the CDAM [see Section III-C, Fig. 3(c)] and the ALFM (see Section III-D, Fig. 5), where SPVM is integrated into CDAM to perform feature extraction.

A. Diff-Attention-Aware State-Space Fusion Model

Given that directly concatenating MS and PAN inputs in a single branch risks disrupting their intrinsic feature structures and amplifying redundancy, the DASF-Model thus employs a dual-branch architecture. This design is essential to empowering our CDAM to separate common features from distinctive ones, thereby eliminating redundancy. The DASF-Model first preprocesses the input MS and PAN images. In the preprocessing block, MS and PAN images undergo upsampling and convolution operations, respectively, to unify their dimensions (unified into $\mathbb{R}^{4\times 64\times 64}$). The upsampling employs bilinear interpolation, while the convolution uses a kernel size of 1 and outputs four channels (equal to the number of spectral bands). Subsequently, a patch operation is applied to transform the visual data into sequential data ($\mathbb{R}^{16\times 16\times 64}$). It should be noted that the sequential data here may still appear to be visual data since it retains three dimensions. However, it can be readily converted into genuine sequential data ($\mathbb{R}^{(16\times 16)\times 64}$) through a simple planar scan operation. The preprocessed results of the two modalities are denoted as MS_p and PAN_p, respectively,

$$\begin{aligned}MS_p &= \text{PreMS}\,(\text{MS})\\ \text{PA}N_p &= \text{PrePAN}\,(\text{PAN})\end{aligned} \tag{5}$$

where $MS_p \in \mathbb{R}^{16\times 16\times 64}$, $\text{PA}N_p \in \mathbb{R}^{16\times 16\times 64}$, and PreMS$(\cdot)$ and PrePAN$(\cdot)$ denote the preprocessing modules for MS and PAN images, respectively.

Then, MS_p and $\text{PA}N_p$ are fed into the CDAM for feature extraction and separation. The CDAM outputs both the common features between MS and PAN (MS_c^1, PAN_c^1) and their respective distinctive features (MS_s^1, PAN_s^1)

$$\left(\text{MS}_c^1, \text{MS}_s^1, \text{PAN}_c^1, \text{PAN}_s^1\right) = \text{CDAM}^1\left(\text{MS}_p, \text{PAN}_p\right) \tag{6}$$

where all output features maintain $\mathbb{R}^{16\times 16\times 64}$. The subscript "$c$" denotes common features, while "s" indicates distinctive features. Superscript numbers (e.g., 1) identify features from the first CDAM.

The common features are further processed by subsequent CDAM for additional extraction and separation, while the distinctive features are channeled into the ALFM module for fusion. At the output of the final CDAM ($\text{CDAM}^n(\cdot)$, n is a hyperparameter), the common features (MS_c^n, PAN_c^n) from both modalities have achieved sufficient similarity. We

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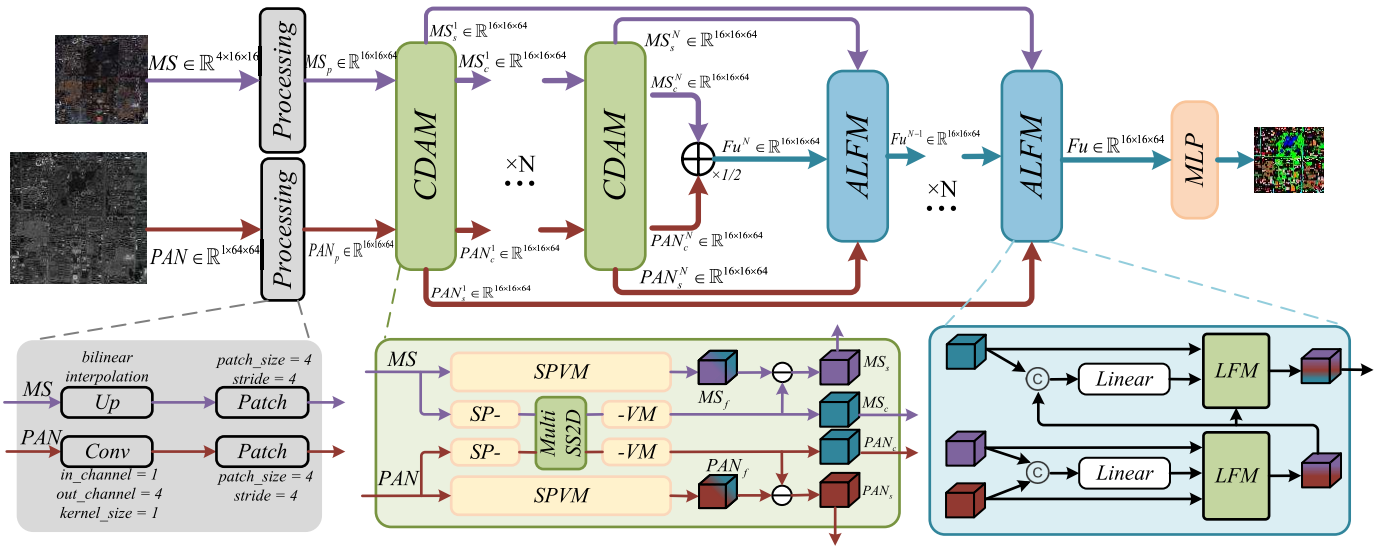


Fig. 2. DASF-Model. CDAM means cross-modal diff-attention module. ALFM is an attention-aware linear fusion module. ⊕ means elementwise sum. N represents the number of CDAM and ALFM.

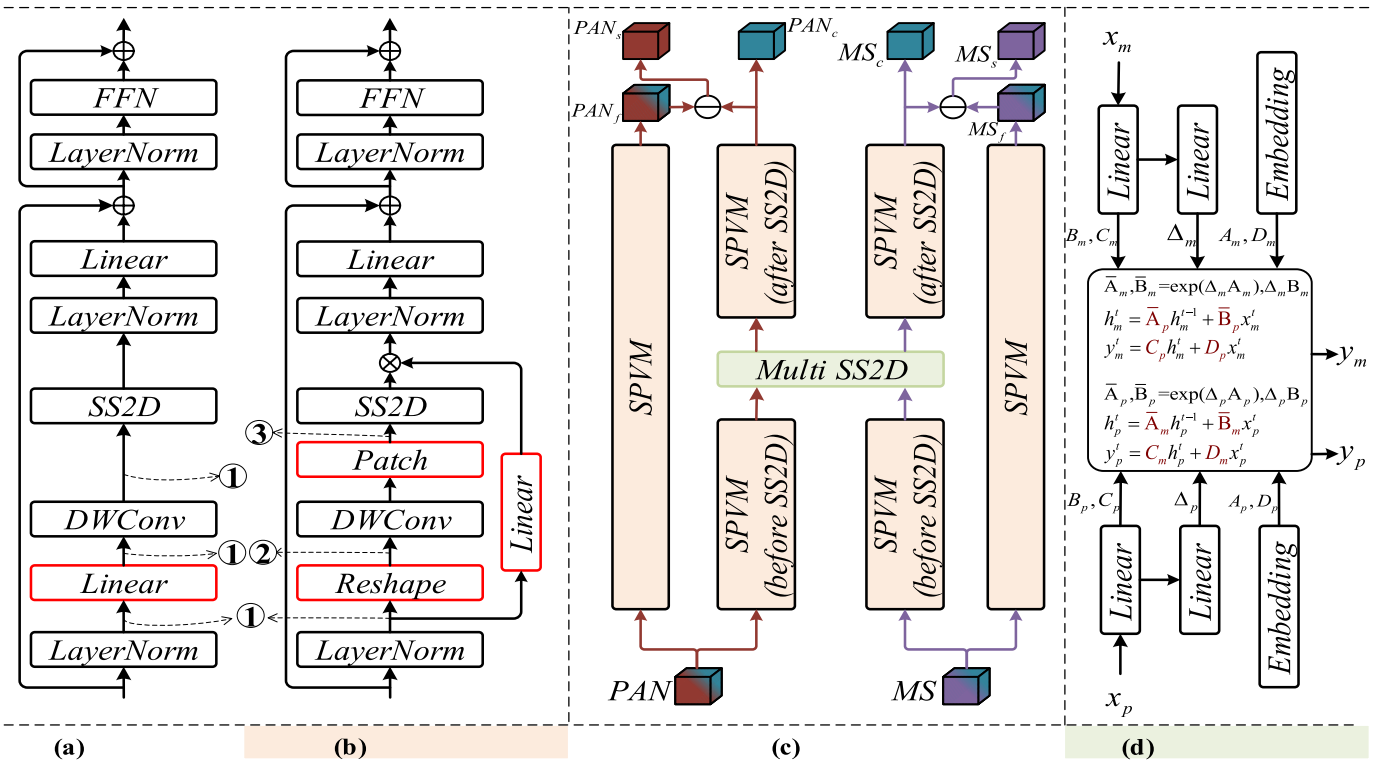


Fig. 3. (a) and (b) Structural designs of VMamba and the proposed SPVM, respectively, along with their comparison. (c) Proposed CDAM, which incorporates the SPVM shown in (b), and (d) Multi-SS2D for acquiring cross-modal shared features. ⊗ means elementwise multiplication. ⊖ represents elementwise subtraction.

therefore perform pixelwise averaging to obtain the fused feature representation $Fu^n \in \mathbb{R}^{16\times16\times64}$

$$\left(\mathrm{MS}_c^n, \mathrm{MS}_s^n, \mathrm{PAN}_c^n, \mathrm{PAN}_s^n\right) = \mathrm{CDAM}^n\left(\mathrm{MS}_c^{n-1}, \mathrm{PAN}_c^{n-1}\right) \quad (7)$$

$$Fu^n = \left(\mathrm{MS}_c^n + \mathrm{PAN}_c^n\right)/2. \quad (8)$$

Subsequently, through a series of ALFM, Fu^n gradually integrates the advantageous features of MS and PAN images, ultimately obtaining the final redundancy-free multimodal fusion feature $\mathrm{Fu} \in \mathbb{R}^{16\times16\times64}$

$$Fu^{n-1} = \mathrm{ALFMM}^n(\mathrm{MS}_s^n, \mathrm{PAN}_s^n, Fu^n) \quad (9)$$

$$\mathrm{Fu} = \mathrm{ALFMM}^1(\mathrm{MS}_s^1, \mathrm{PAN}_s^1, Fu^1). \quad (10)$$

Finally, it is directly entered into a multilayer perceptron (MLP) for subsequent classification

$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{ij}\log(\mathrm{MLP}(\mathrm{Fu}))_{ij}. \quad (11)$$

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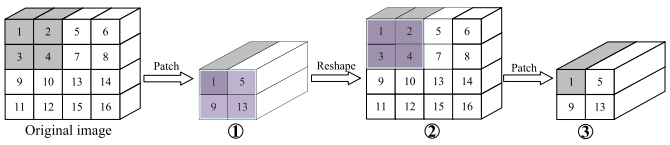


Fig. 4. Specific presentation of related features in Fig. 3(a) and (b).

Among them, N is the number of samples in the batch, C is the number of labeled categories in the ground truth, y_{ij} indicates whether the ith sample belongs to the jth class or not, and its value is 0 or 1. $(M_{lp}(\mathrm{Fu}))_{ij}$ denotes the probability that the ith sample belongs to the jth class.

The DASF-Model cleverly separates the common and dominant features of MS and PAN images through the cross-modal diff-attention mechanism. It is particularly worth mentioning that the ALFM effectively incorporates features with large semantic differences by using influence coefficients to guide pixel-by-pixel fusion, which further improves the robustness and generalization ability of the model.

B. Space Preserving Visual Mamba

In Visual Mamba [see Fig. 3(a)], the 1-D convolution in Mamba is replaced with a DWConv. However, since the input image is first partitioned into nonoverlapping patches and processed as sequential data (see Fig. 4 ①), the convolved regions in DWConv may no longer correspond to spatially adjacent areas in the original image. This design not only fails to effectively capture local features but also introduces unnecessary spatial redundancy. To address this limitation, we propose the SPVM [see Fig. 3(b)], which efficiently extracts local features while modeling contextual relationships between independent patches.

1) Extracting Local Features With Contextual Information: To ensure convolutional operations are performed on spatially adjacent regions in the original image, we reshape the features ($\mathbb{R}^{16\times16\times64}$) prior to DWConv. Specifically, we first partition each patch along the channel dimension ($\mathbb{R}^{16\times16\times(4\times4\times4)}$), then reassemble the separated patches back to their original spatial positions ($\mathbb{R}^{(16\times4)\times(16\times4)\times4}$). Finally, the features are reshaped into 3-D form ($\mathbb{R}^{4\times64\times64}$, Fig. 4 ②). This processed feature representation enables DWConv to effectively capture both local features and interpatch contextual relationships. Notably, the linear layer in Visual Mamba employs an expansion ratio of 1, rendering its inclusion questionable. This layer not only introduces unnecessary computational overhead but may also interfere with the transformation between 2-D sequential and 3-D image representations. Consequently, we eliminate this linear layer in our design

$$F_2 = \mathrm{Reshape}\left(F_1\right). \tag{12}$$

2) Extracting Global Features From Sequences: Since Mamba's scanning operation is inherently sequential, it is not directly applicable to visual data. To address this, Visual Mamba introduced the 2-D selective scan (SS2D) block. In our work, we adopt this SS2D block for feature processing. Prior to this operation, we first repatch the features to transform them back into sequential form (Fig. 4 ③)

$$F_3 = \mathrm{Patch}\left(\mathrm{DWConv}\left(F_2\right)\right) \tag{13}$$

in which, Patch(·) is implemented by a convolutional layer, where both the kernel size and stride are set to 4 (consistent with the patch size).

The sequence is then processed by the SS2D block for global modeling, followed by output linear projection and feed-forward network, ultimately yielding the output features $F_{\mathrm{out}} \in \mathbb{R}^{16\times16\times64}$ of our SPVM. We additionally maintain the multiplicative branch z from Mamba (omitted in Visual Mamba), which helps the model to better attend to downstream-relevant key information

$$z = \mathrm{Linear}\left(F_1\right), \tag{14}$$

$$F_{\mathrm{out}} = \mathrm{FNN}\left(\mathrm{Linear}\left(\mathrm{SS2D}\left(F_3\right) \times z\right)\right). \tag{15}$$

There exists a fundamental difference between computer vision methods and SSMs: traditional vision methods typically process 3-D tensor data, while SSMs, as sequence models, require 2-D sequential inputs. This dimensional difference hinders the effective combination of their respective advantages. Our SPVM achieves more flexible input reshaping and patching operations by removing the input linear mapping layer in the Visual Mamba architecture and adjusting feature dimensions. It not only seamlessly combines with traditional vision methods but also creates conditions for deeper integration between sequence modeling and visual processing paradigms.

C. Cross-Modal Diff-Attention Module

MS and PAN are RS images of different modalities for the same scene, which means they contain substantial semantic redundancy. To address this, we propose a CDAM to reduce redundant information while improving feature representation. The detailed implementation is shown in Fig. 3(c).

1) Extracting Intramodal Features: Here, we employ two parameter-independent SPVM blocks to process the input features ($\mathrm{MS_p}$, $\mathrm{PAN_p}$) of MS and PAN separately, obtaining all intramodal features (MS_f^1, PAN_f^1) for each modality

$$\mathrm{MS}_f^1 = \mathrm{SPVM}\left(\mathrm{MS}_p\right) \tag{16}$$

$$\mathrm{PAN}_f^1 = \mathrm{SPVM}\left(\mathrm{PAN}_p\right) \tag{17}$$

where MS_f^1 and PAN_f^1 represent all features within the MS and PAN modalities, respectively, encompassing both modality-specific advantageous features and common features.

2) Extraction Intermodal Common Essential Feature: The SSM produces learnable parameter matrices (A, B, C, D, Δ) during feature processing, with each matrix specializing in different extraction stages to guide modality-specific salient feature learning. A recent study [35] has demonstrated that these matrices function similar to the QKV matrices in Transformer architectures.

From a signal processing point of view, the response between signals with similar feature distributions is stronger than that between signals with different feature distributions.

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That is, the response of the common part between the multimodal features is stronger than the response of the respective advantage part.

Inspired by this, we design a Multi-SS2D block to extract cross-modal common features. Within Multi-SS2D, we process MS/PAN features using parameter matrices generated from PAN/MS. Specifically,

$$h_m^t = \bar{\mathrm{A}}_p h_m^{t-1} + \bar{\mathrm{B}}_p x_m^t$$
$$y_m^t = C_p h_m^t + D_p x_m^t \tag{18}$$
$$h_p^t = \bar{A}_m h_p^{t-1} + \bar{B}_m x_p^t$$
$$y_p^t = C_m h_p^t + D_m x_p^t \tag{19}$$

where A, B, C, D, and Δ are SSM parameter matrices, with h as hidden state and x and y as input or output sequences. Subscripts m/p denote MS/PAN modalities. Superscripts $t/t-1$ indicate current/previous time steps.

Subsequently, we replace the SS2D block in SPVM with Multi-SS2D while maintaining identical designs for all other blocks. For notational convenience, we denote the portion before SS2D as $\mathrm{SPVM}_b(\cdot)$ and the portion after SS2D as $\mathrm{SPVM}_a(\cdot)$

$$\mathrm{MS}_c^1 = \mathrm{SPV}M_a\left(\mathrm{MSS2}D\left(\mathrm{SPV}M_b\left(MS_p\right)\right)\right)$$
$$\mathrm{PAN}_c^1 = \mathrm{SPV}M_a\left(\mathrm{MSS2}D\left(\mathrm{SPV}M_b\left(\mathrm{PA}N_p\right)\right)\right) \tag{20}$$

where $\mathrm{MSS2}D(\cdot)$ denotes the Multi-SS2D and MS_c^1 and PAN_c^1 represent the cross-modal common features in MS and PAN modalities, respectively. The subscript c denotes common features, and the superscript 1 signifies that this is the output from the first CDAM.

3) Obtain Respective Advantageous Features: Intramodal feature MS_f^1 and PAN_f^1 contains all the information of its corresponding branch. By subtracting these intermodal common essential features from the intramodal features of the branch, the advantageous features of the MS and PAN modality MS_s^1 and PAN_s^1 can be obtained, respectively, specifically,

$$\mathrm{MS}_s^1 = \mathrm{MS}_f^1 - \mathrm{MS}_c^1 \tag{21}$$
$$\mathrm{PAN}_s^1 = \mathrm{PAN}_f^1 - \mathrm{PAN}_c^1. \tag{22}$$

The CDAM can effectively separate the common essential features and respective advantageous features of MS and PAN images, which significantly reduces redundancy in subsequent feature fusion while enhancing the model's robustness and generalization capability.

D. Attention-Aware Linear Fusion Module

Considering that the features to be fused are the respective advantages of MS and PAN and their common essential features, there are considerable differences. This article proposes an ALFM. This module can achieve dimensionally stable fusion across features with substantial semantic disparities, see Fig. 5 for details.

For the three input features $((\mathrm{MS}_s, \mathrm{PAN}_s, Fu^N) \in \mathbb{R}^{16\times16\times64})$, we fuse MS_s and PAN_s to obtain the fused features $F_{\mathrm{mp}} \in \mathbb{R}^{16\times16\times64}$ first. Then F_{mp} is fused with Fu^N to produce the fusion result Fu^{N-1}. The following takes the fusion of MS_s and PAN_s as an example to detail the fusion process.

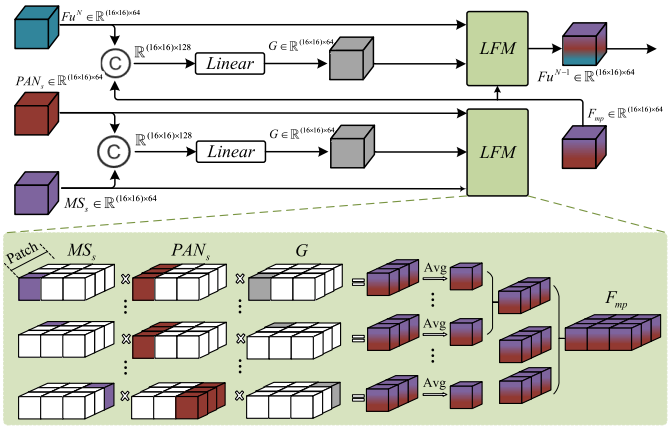


Fig. 5. ALFM. LFM is a linear fusion operation. circledC denotes concatenation along the last dimension. G represents the pixel-by-pixel attention coefficient matrix generated by the fused features.

1) Generate Attention Coefficient Matrix G: First, MS_s and PAN_s are unfolded into sequential representations ($\mathbb{R}^{(16\times16)\times64}$), where 16×16 indicates 256 patches in total, and 64 denotes 64 pixels per patch. These are then concatenated along the last dimension and processed through a linear projection to obtain the pixelwise attention coefficient matrix $G \in \mathbb{R}^{(16\times16)\times64}$. Each value in G represents the influence coefficient of PAN_s on the corresponding position in MS_s. This pixelwise attention coefficient independently guides the fusion of each pixel, significantly alleviating the issue of large discrepancies in fused features

$$G = \mathrm{Linear}\left([\mathrm{MS}_s, \mathrm{PAN}_s]\right). \tag{23}$$

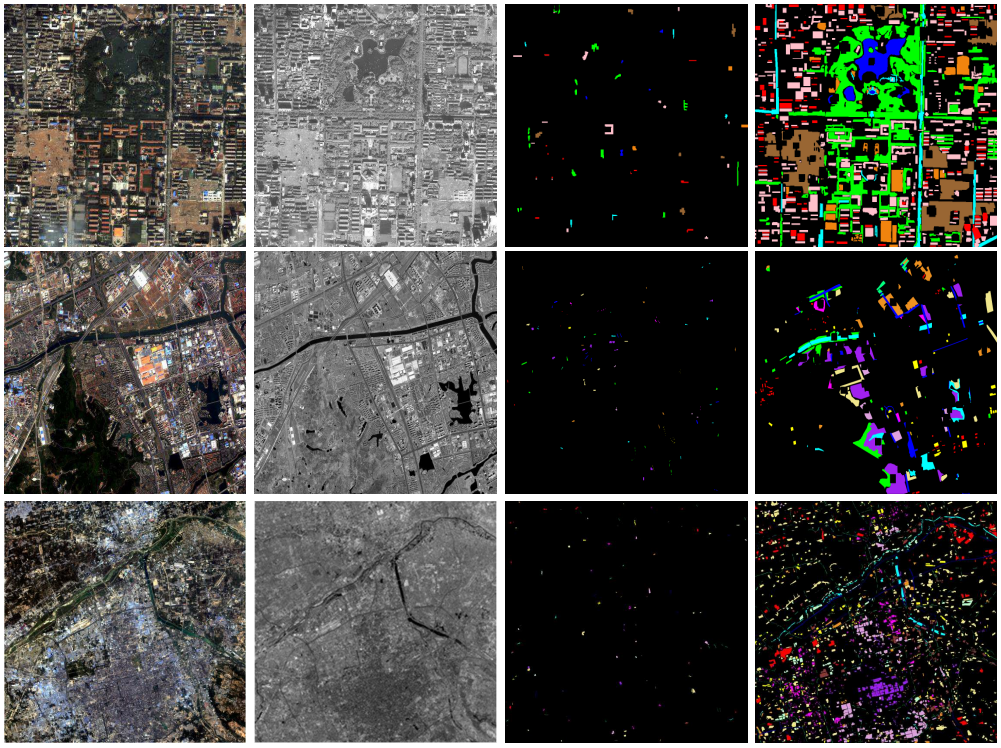
2) Get the Fusion Result: After deriving G, both MS_s and PAN_s, along with G, are fed into the linear fusion module (LFM). In LFM, each feature of every patch in MS_s is multiplied by the corresponding patch in PAN_s, which can be interpreted as the influence of PAN_s on that specific feature in MS_s. Subsequently, under the guidance of its corresponding attention coefficient from G, the feature fusion matrix is computed. Finally, global average pooling is applied to obtain the value of that feature in F_{mp}. By repeating this linear fusion process for all features in MS_s, we obtain the fused feature F_{mp}

$$F_{\mathrm{mp}} = \mathrm{LFM}\left(\mathrm{MS}_s, \mathrm{PAN}_s, G\right). \tag{24}$$

The ALFM performs featurewise fusion guided by computed influence coefficients. So, it can show good results when fusing features with large differences, which greatly improves the utilization of multimodal image and the generalization ability of models.

The DASF-Model (pseudocode provided in Algorithm 1) separates the common essential features and their respective dominant features of multimodal image by the CDAM. Among this, the SPVM extracts both local and global features of the input image. Moreover, the ALFM is used to fuse them to further improve the classification performance and generalization ability of the model.

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Color	Class	Train	Test
	Building	1744	69985
	Flatland	560	17867
	Road	523	15533
	Shadow	665	20367
	Soil	1367	49483
	Tree	2518	100492
	Water	397	12967
	Water	1600	58236
	Road1	1858	54239
	Bareland	1114	43403
	Vegetation	2012	61715
	Red Building	1663	53864
	Blue Building	272	12054
	Intensive Building	3467	126012
	White Building	604	17621
	Sparse Building	530	15397
	Road2	254	8583
	Lowvegetation	644	11737
	Water	2676	125251
	Road1	1558	65447
	Road2	1817	96698
	Bareland	2566	116160
	Lowvegetation	3617	157373
	Farmland	5578	255036
	Tree	4175	189697
	Intensive Building	4086	169072
	Sparse Building	8309	341197
	Suburb Building	14055	581470
	Workshop Building	4018	165888
	White Building	3437	145348

Fig. 6. Visualization of datasets. (Top to bottom) Scenes of Xi'an Urban, Nanjing, and Xi'an. (Left to right) MS images, PAN images, training sets, testing sets, and related data statistics.

Algorithm 1 DASF-Model

Require: MS: the MS image; PAN: the PAN image; y: the ground-truth of MS image;
Ensure: P_{model}: trained model;

```
epoch = 0;
while epoch < max _epoch do
    MS_p = PreMs(MS) (Eq. (5));
    PAN_p = PrePAN(PAN) (Eq. (5));
    while i < N do
        (MS_c^i, MS_s^i, PAN_c^i, PAN_s^i) = CDAM^i(MS_c^{i-1}, PAN_c^{i-1})
        (Sec. III-C);
    end while
    Fu^n = (MS_c^n + PAN_c^n)/2 (Eq. (8));
    while i < N do
        Fu^{i-1} = ALFM^i(Fu^i, MS_s^i, PAN_s^i) (Sec. III-D);
    end while
    L = -(1/N) Σ_{i=1}^{N} Σ_{j=1}^{C} y_ij log(MLP(Fu))_ij (Eq. (11));
    epoch = epoch + 1;
end while
```

IV. Experimental Study

This section systematically evaluates the DASF-Model's performance through four key aspects: datasets and equipment specifications, hyperparameter selection, ablation experiment, and feature visualization and comparative analysis with existing methods.

A. Datasets Description and Experimental Settings

1) Datasets: This article utilizes four paired MS and PAN datasets, three of which (see Fig. 6) are obtained from the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, while the last one (see Fig. 7) is a publicly available dataset. The datasets are described in Table I.

2) Experimental Settings: The experiments conducted in this study were performed on a computing platform equipped with an RTX2080Ti GPU (12-GB memory) and 128-GB RAM. The network is trained through the use of the PyTorch framework. Throughout the training phase, the weight decay is established at 0.0005, with the number of epochs set to 25, and the initial learning rate for Xi'an dataset is 0.000393, while the other two datasets use a learning rate of 0.000293. Meanwhile, the learning rate decays to 0.9995 times its previous value after each training batch. For Nanjing and Xi'an datasets, a batch size of 128 is implemented, whereas Xi'an Urban datasets are trained using a batch size of 64.

3) Quality Evaluation Indicators: Our experimental protocol incorporates a multidimensional evaluation framework to rigorously assess model performance. The assessment methodology examines three principal metrics: overall accuracy (OA) serves as the primary indicator of global prediction correctness across all classes; average accuracy (AA) provides insights into potential classification bias by evaluating mean performance distribution among categories; Kappa coefficient statistic (Kappa) offers a chance-corrected agreement measurement that accounts for random prediction effects, delivering more statistically robust performance interpretation. This comprehensive metric system ensures thorough validation of the generalization capacity and operational stability of the DASF-Model under diverse conditions.

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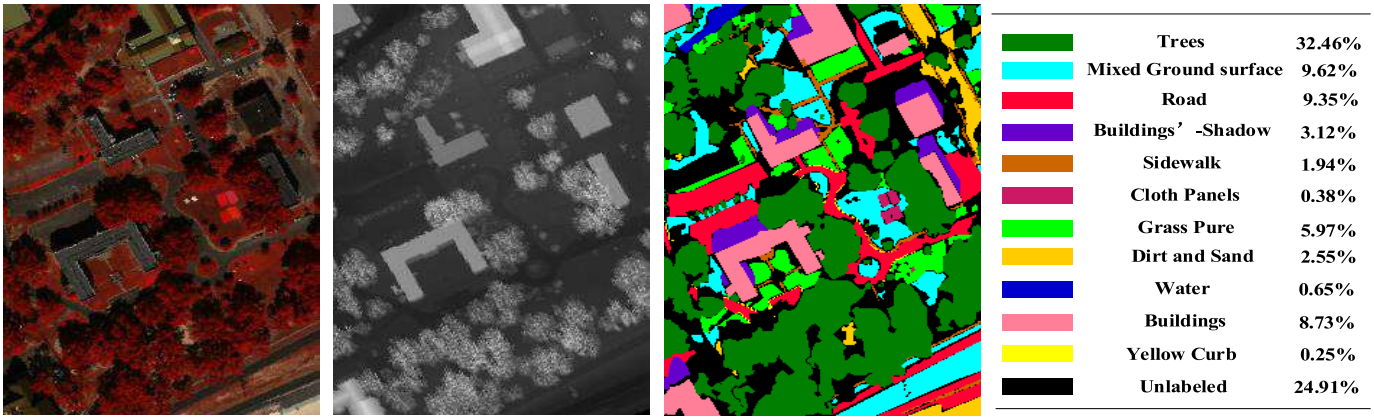


Fig. 7. Visualization of MUUFL dataset. (Left to right) MS images, PAN images, ground truth (GT), and related data statistics.

TABLE I
RELEVANT DETAILS OF THE EMPLOYED DATASETS

Dataset	MS			PAN			Classes	Total Samples	Training Samples	Test Samples
	Size	Bands	Resolution	Size	Bands	Resolution				
Xi'an Urban	800× 830	4	8m	2400× 2520	1	2m	7	294468	7774	286694
Nanjing	2000× 2500	4	8m	8000× 10000	1	2m	11	476879	14018	462863
Xi'an	4548× 4541	4	8m	18192× 18164	1	2m	12	2364529	55892	2408637
Muufl	325× 220	64	1m	325× 220	2	1m	11	53678	2699	50988

TABLE II
EXPERIMENTAL RESULT OF THE DASF-MODEL HYPERPARAMENTER BLOCK NUMBERS *N*

Dataset	Xi'an Urban				Nanjing				Xi'an			
N	[1,1]	[2,2]	[3,3]	[4,4]	[1,1]	[2,2]	[3,3]	[4,4]	[1,1]	[2,2]	[3,3]	[4,4]
OA	**85.35**	83.91	83.38	82.41	77.35	**79.11**	76.32	74.29	80.50	80.38	**80.56**	79.29
AA	**79.20**	77.65	73.35	75.72	73.72	**73.92**	71.87	69.90	73.90	73.63	**73.93**	72.13
Kappa	**81.14**	79.33	78.39	77.39	73.70	**75.54**	72.40	70.09	77.69	77.54	**77.74**	76.31

B. Selection of Hyperparameters

Since the feature separation and fusion operations exhibit a one-to-one correspondence, the number of CDAM and ALFM should remain consistent, denoted as N. In order to explore the influence of N on the DASF-Model, we set N to 1–4 to compare the accuracy versus the parameters of the three datasets.

The results are shown in Table II. On the datasets of Xi'an Urban, Nanjing, and Xi'an, the network attains the best accuracy when N is 1–3, respectively. This is because datasets of varying sizes require different numbers of modules to effectively separate the common and dominant features across the two modalities. For the smallest Xi'an Urban dataset, only one CDAM is needed to complete the separation of common features and dominant features. However, for the largest Xi'an dataset, three CDAMs are needed to complete. Therefore, when extending to new datasets, only the value of N needs to be adjusted according to the dataset size—larger datasets use a bigger N while smaller datasets use a smaller N. In each dataset, when the value of N exceeds the optimal threshold, the performance of the model will be decreased, because when N is equal to the optimal threshold, the common features and the dominant features are separated, and the subsequent excessive modules produce more redundant features in the dominant features, resulting in the failure of the feature separation module, and then the performance of the model will be reduced.

C. Ablation Experiment

In this article, we have chosen all four datasets for a comprehensive combinatorial investigation involving SPVM, CDAM, as well as ALFM. By conducting experiments with various combinations of these modules, we aim to assess their individual effectiveness and necessity within the network architecture. The detailed results of these experiments are meticulously documented in Tables III–VI, providing insights into the impact and significance of each module in optimizing network performance. Fig. 8 shows the classification results under five experimental settings on Xi'an Urban dataset. Next, we take the results of Xi'an Urban dataset as an example for analysis.

In the experimental verification on Xi'an Urban dataset, experiment (1) used simple pixelwise averaging to fuse bimodal features, while experiment (2) employed the proposed

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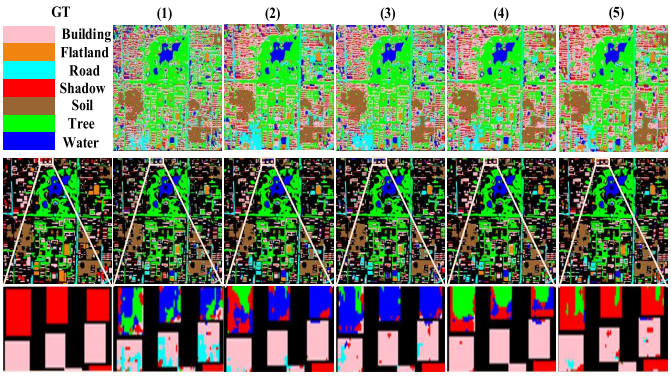


Fig. 8. Ablation study visualization results on Xi'an Urban dataset. The leftmost column contains the legend and GT.

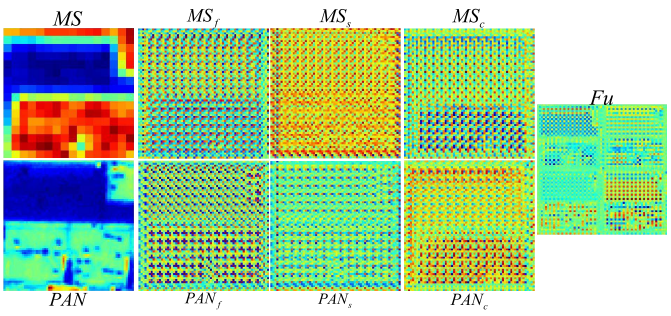


Fig. 9. Visualization results of intermediate feature maps. The first row displays MS features, while the second row shows PAN features. (Left to right) Five columns represent input images, intramodal features, advantageous features, common features, and fused features, respectively.

TABLE III
RESULTS OF THE ABLATION STUDY ON XI'AN URBAN IMAGE

	Visual Mamba	CDAM	SPVM	ALFM	OA(%)	AA(%)	Kappa(%)
(1)	√				79.02	71.01	72.92
(2)	√			√	82.25	75.30	77.23
(3)	√	√			80.81	72.62	75.25
(4)	√	√		√	83.77	76.48	79.01
(5)		√	√	√	**85.35**	**79.20**	**81.14**

TABLE IV
RESULTS OF THE ABLATION STUDY ON XI'AN IMAGE

	Visual Mamba	CDAM	SPVM	ALFM	OA(%)	AA(%)	Kappa(%)
(1)	√				77.40	70.02	74.58
(2)	√			√	79.01	71.86	76.19
(3)	√	√			78.29	71.05	75.47
(4)	√	√		√	79.77	72.57	76.95
(5)		√	√	√	**80.56**	**73.93**	**77.74**

TABLE V
RESULTS OF THE ABLATION STUDY ON NANJING IMAGE

	Visual Mamba	CDAM	SPVM	ALFM	OA(%)	AA(%)	Kappa(%)
(1)	√				75.95	70.54	72.48
(2)	√			√	77.56	71.89	73.99
(3)	√	√			76.84	71.07	73.27
(4)	√	√		√	78.32	72.56	74.75
(5)		√	√	√	**79.11**	**73.92**	**75.54**

TABLE VI
RESULTS OF THE ABLATION STUDY ON THE MUUFL IMAGE

	Visual Mamba	CDAM	SPVM	ALFM	OA(%)	AA(%)	Kappa(%)
(1)	√				92.51	81.50	90.06
(2)	√			√	93.58	80.97	91.49
(3)	√	√			92.93	81.96	90.65
(4)	√	√		√	93.88	82.65	91.90
(5)		√	√	√	**95.09**	**84.84**	**93.50**

ALFM module for fusion. The results show that the ALFM module significantly outperforms naive averaging fusion, with a substantial boost in classification accuracy. Experiment (3) introduced the proposed CDAM based on experiment (1) to separate shared and distinct advantageous features between modalities while reducing redundancy. Experimental results confirm that the CDAM effectively enhances classification performance. Experiment (4) further combined CDAM and ALFM to validate their synergy. The results demonstrate that the two modules exhibit strong compatibility and synergistic effects without mutual performance interference. Experiment (5) replaced the Visual Mamba backbone with the proposed SPVM in the framework of experiment (4). Experiments prove that SPVM achieves significant improvements across all three evaluation metrics compared with Visual Mamba, further elevating overall model performance.

D. Feature Visualization

To demonstrate that the feature separation and redundancy removal process operates as intended, this section presents visualizations of intermediate feature maps (Fig. 9). It can be observed that the intramodal features of MS are predominantly red and blue, while those of PAN are mainly cyan and blue, which aligns with their respective input images. Therefore, the advantageous features of MS and PAN should correspond to red and cyan features, respectively, while their common features should primarily be blue.

The third column displays the advantageous features, which indeed match our expectations. The fourth column presents the common features, both dominated by blue. However, the common features of MS contain more cyan, while those of PAN include more red. This occurs because they are processed using the parameter matrices from the opposite modality. The last column presents the fused features generated by our ALFM method, demonstrating that the fusion result effectively integrates both the common features of MS and PAN and their respective advantageous features.

E. Comparison Algorithms and Experiment Results

In Xi'an Urban, Nanjing, Xi'an, and Muufl dataset, the proposed DASF-Model is compared with eight state-of-the-art methods, namely, CRHFF [32], AM^{3}Net [36], GCFNet [37], MFT [25], DFINet [33], NNCNet [34], 3DSADiff [38], and ISSP [23]. Tables VII–X show the results of the quantitative

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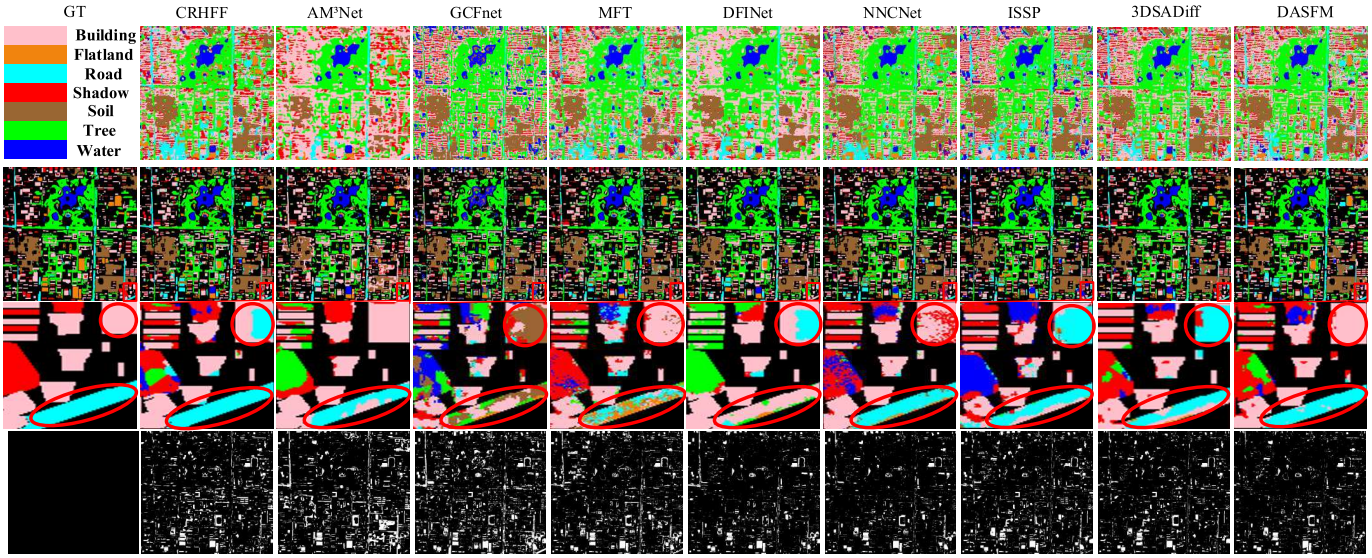


Fig. 10. Visualization results of the experiment on Xi'an Urban. The leftmost column contains the legend and GT. Each subsequent column contains an overall color map and a label position color map. (Left to right) GT, CRHFF, AM^{3}Net, GCFNet, MFT, DFINet, NNCNet, ISSP, 3DSADiff, and DASF-Model.

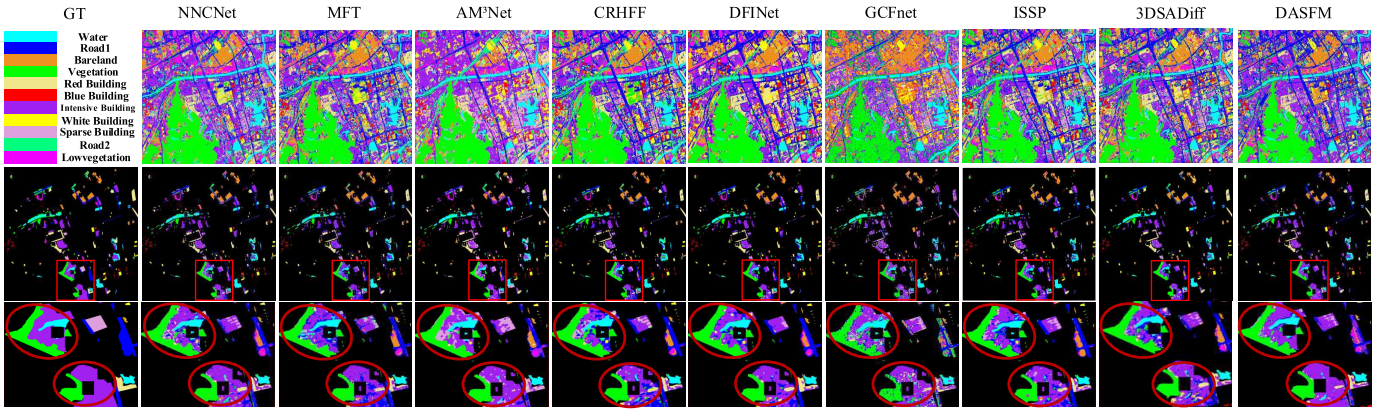


Fig. 11. Visualization results of the experiment on Nanjing. The leftmost column contains the legend and GT. Each subsequent column contains an overall color map and a label position color map. (Left to right) GT, NNCNet, MFT, AM^{3}Net, CRHFF, DFINet, GCFNet, ISSP, 3DSADiff, and DASF-Model.

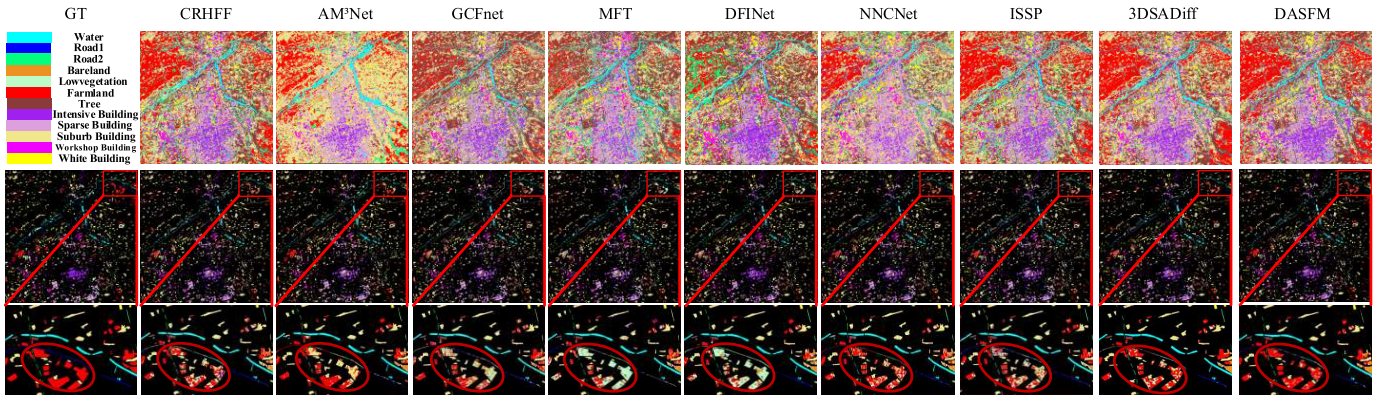


Fig. 12. Visualization results of the experiment on Xi'an. The leftmost column contains the legend and GT. Each subsequent column contains an overall color map and a label position color map. (Left to right) GT, CRHFF, AM^{3}Net, GCFNet, MFT, DFINet, NNCNet, ISSP, 3DSADiff, and DASF-Model.

comparison experiments, respectively, and Figs. 10–13 show the visualizations of the comparison algorithms, respectively. All quantitative results presented are the average of five experiments. All comparative results were obtained by tuning each algorithm to its optimal hyperparameter configuration on the respective dataset.

The CRHFF utilizes integrated shallow-to-deep features combined with global-to-local characteristics from

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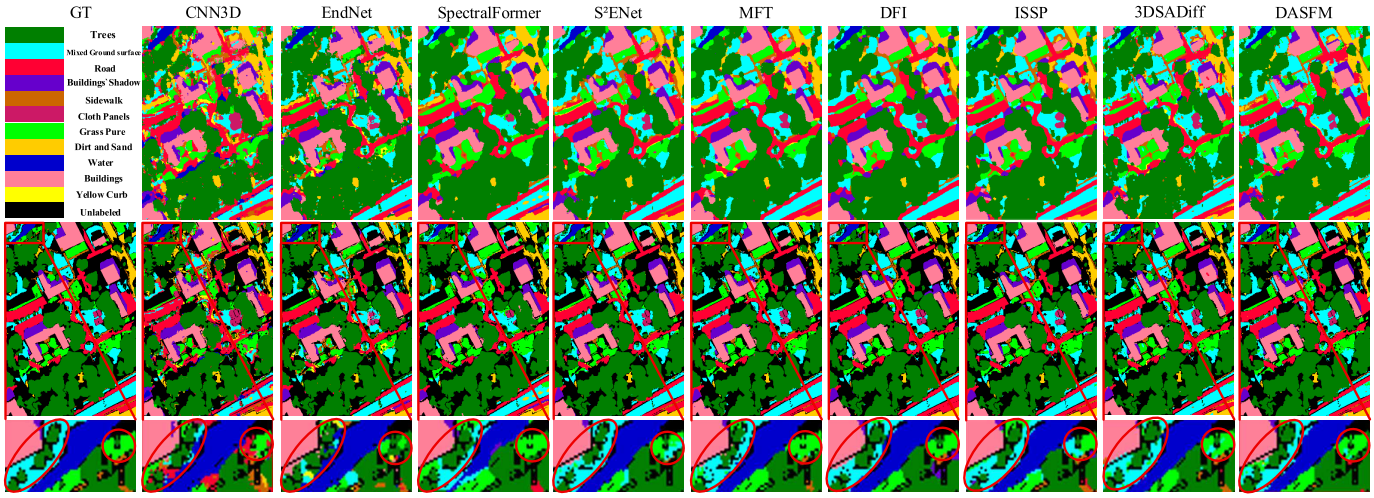


Fig. 13. Visualization results of the experiment on MUUFL. The leftmost column contains the legend and GT. Each subsequent column contains an overall color map and a label position color map. (Left to right) GT, CNN3D, EndNet, SpectralFormer, S^{2}ENet, MFT, DFINet, ISSP, 3DSADiff, and DASF-Model.

TABLE VII

QUANTITATIVE RESULTS OF XI'AN URBAN DATASET. THE BEST THREE RESULTS FOR EACH CATEGORY ARE HIGHLIGHTED IN RED, BLUE, AND BROWN, RESPECTIVELY

Models	CRHFF [32]	AM^{3}Net [36]	GCFNet [37]	MFT [25]	DFINet [33]	NNCNet [34]	ISSP [23]	3DSADiff [38]	DASF-Model
c_1: Building	77.53	84.27	72.96	74.96	71.58	85.62	76.39	84.11	84.23
c_2: Flatland	40.93	35.89	30.75	46.82	52.87	38.35	37.21	29.53	36.77
c_3: Road	73.86	39.09	25.46	43.74	80.65	59.89	64.04	72.39	74.04
c_4: Shadow	75.48	58.12	42.02	68.17	45.33	71.41	83.35	40.36	79.21
c_5: Soil	90.99	55.32	80.74	81.70	92.98	90.42	89.72	90.35	90.76
c_6: Tree	79.98	82.43	88.36	82.21	85.55	92.55	83.22	91.40	93.78
c_7: Water	98.04	96.86	71.37	98.09	98.71	95.61	97.89	93.72	98.58
OA(%)	79.01	71.87	72.23	75.78	78.86	83.98	79.44	82.14	85.35
AA(%)	76.69	64.57	58.81	70.81	75.38	76.26	75.97	72.52	79.20
Kappa(%)	73.35	63.19	63.97	68.89	72.82	79.31	73.80	77.03	81.14
Time(s)	127.99	176.56	268.79	17.63	49.72	88.15	238.15	51.84	40.36
Parameter(n)	4906823	2777838	3878187	212876	759438	2212423	12082537	19604357	498035

TABLE VIII

QUANTITATIVE RESULTS OF NANJING DATASET. THE BEST THREE RESULTS FOR EACH CATEGORY ARE HIGHLIGHTED IN RED, BLUE, AND BROWN, RESPECTIVELY

Models	CRHFF [32]	AM^{3}Net [36]	GCFNet [37]	MFT [25]	DFINet [33]	NNCNet [34]	ISSP [23]	3DSADiff [38]	DASF-Model
c_1: Water	94.86	88.44	93.45	94.23	96.18	94.77	94.24	94.99	96.02
c_2: Road1	72.96	40.62	37.36	64.21	72.09	67.25	59.47	67.37	67.73
c_3: Bareland	94.24	54.10	84.98	91.84	92.23	88.19	94.50	87.23	93.95
c_4: Vegetation	84.85	71.55	78.65	76.58	88.77	75.83	80.80	86.80	79.93
c_5: Red Building	77.61	67.85	24.98	75.54	85.50	77.94	78.04	79.94	79.95
c_6: Blue Building	95.59	93.66	44.95	91.60	97.09	93.03	75.33	94.53	95.32
c_7: Intensive Building	59.08	74.96	64.42	64.48	64.35	75.72	69.34	68.23	77.07
c_8: White Building	81.76	90.98	64.74	75.73	86.71	84.37	58.44	84.96	85.44
c_9: Sparse Building	53.61	80.99	45.62	38.92	44.81	47.68	52.80	50.47	44.50
c_{10}: Road2	43.07	91.86	13.58	34.21	30.92	53.03	33.90	54.71	34.57
c_{11}: Lowvegetation	62.74	43.96	47.84	63.27	63.27	66.78	59.09	54.66	58.64
OA(%)	75.52	70.22	61.65	73.35	78.00	77.77	75.01	76.98	79.11
AA(%)	74.85	72.63	54.60	70.06	74.72	74.96	70.54	74.90	73.92
Kappa(%)	71.77	65.02	55.01	69.11	74.47	74.03	71.03	73.31	75.54
Time(s)	206.27	140.97	198.96	28.57	79.11	142.89	286.71	121.29	118.25
Parameter(n)	4906823	2777838	3878187	212876	759438	2212423	12082537	19604357	821803

high-resolution MS and PAN images. By extracting and fusing cross-resolution information, it enhances multimodal representation modeling, leading to improved classification accuracy. The AM^{3}Net can measure the contribution rate of each pixel's spectral channel to the construction of spectral feature tensor independently and realize the cooperative transmission of spectral features and spatial information through a spectral spatial mutual conduction module. The GCFNet is an encoder–decoder-based global cooperative fusion network, which improves performance by exploiting the context dependency and fusing sampled and unsampled location data. In addition, an adaptive loss-weighted fusion

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TABLE IX

QUANTITATIVE RESULTS OF XI'AN DATASET. THE BEST THREE RESULTS FOR EACH CATEGORY ARE HIGHLIGHTED IN RED, BLUE, AND BROWN, RESPECTIVELY

Models	CRHFF [32]	AM^{3}Net [36]	GCFNet [37]	MFT [25]	DFINet [33]	NNCNet [34]	ISSP [23]	3DSADiff [38]	DASF-Model
c_1: Water	95.89	95.61	95.89	97.35	93.51	93.15	96.75	96.74	96.62
c_2: Road1	18.35	1.72	17.46	16.82	15.41	34.16	26.29	19.60	22.26
c_3: Road2	62.01	11.61	55.38	50.69	53.66	62.14	68.75	75.26	62.95
c_4: Bareland	66.45	67.74	66.46	64.58	75.55	69.01	72.70	70.60	69.47
c_5: Lowvegetation	55.51	18.81	36.06	46.68	45.00	55.10	54.81	46.86	45.02
c_6: Farmland	69.74	76.35	64.31	51.69	50.59	68.41	71.42	72.44	73.43
c_7: Tree	84.84	37.73	89.12	91.78	92.29	77.92	86.47	87.90	90.44
c_8: Intensive Building	77.59	72.44	72.65	62.85	86.75	74.62	74.57	79.29	83.43
c_9: Sparse Building	77.18	80.85	76.40	75.73	78.26	77.48	77.17	75.25	75.59
c_{10}: Suburb Building	95.51	99.81	97.81	91.90	96.06	96.81	95.47	98.35	97.67
c_{11}: Workshop Building	95.39	75.48	90.71	94.52	96.75	94.23	92.03	93.51	93.79
c_{12}: White Building	75.33	45.00	66.36	72.87	83.25	76.02	67.29	70.88	76.43
OA(%)	79.42	69.58	76.86	74.65	78.70	79.24	79.58	80.33	80.56
AA(%)	72.81	56.93	69.05	68.12	72.26	73.25	73.64	73.89	73.93
Kappa(%)	76.44	64.18	73.48	71.08	75.78	76.24	76.66	77.49	77.74
Time(s)	1159.44	1041.57	2474.43	140.23	425.58	756.16	872.85	336.04	654.26
Parameter(n)	4906823	2777838	3878187	212876	759438	2212423	12082537	19604357	1140411

TABLE X

QUANTITATIVE RESULTS OF THE MUUFL DATASET. THE BEST THREE RESULTS FOR EACH CATEGORY ARE HIGHLIGHTED IN RED, BLUE, AND BROWN, RESPECTIVELY

Models	CNN3D [39]	EndNet [40]	SpectralFormer [41]	S^{2}ENet [26]	MFT [25]	DFINet [33]	ISSP [23]	3DSADiff [38]	DASF-Model
c_1: Trees	95.10	95.16	97.30	98.42	97.90	98.21	98.06	98.47	98.04
c_2: Mixed Ground Surface	63.72	75.25	69.35	90.75	92.11	89.17	88.23	91.16	89.55
c_3: Road	69.94	81.88	78.48	83.51	91.80	89.85	91.01	91.43	93.69
c_4: Buildings'Shadow	63.90	86.23	82.63	95.38	91.59	91.98	95.30	93.52	93.80
c_5: Sidewalk	79.48	90.97	87.91	95.57	95.60	96.69	96.92	95.52	97.21
c_6: Cloth-Panels	02.86	96.66	58.77	85.77	88.19	96.61	96.00	94.47	98.12
c_7: Grass-Pure	47.96	83.64	85.87	90.15	90.27	92.78	91.53	91.42	92.13
c_8: Dirt-and-Sand	70.47	93.51	95.60	98.22	97.26	98.21	97.78	97.50	97.24
c_9: Water	06.28	60.08	53.52	69.30	61.35	71.96	75.81	68.65	66.69
c_{10}: Buildings	00.00	19.75	08.43	31.60	17.43	16.66	05.98	34.96	25.15
c_{11}: Yellow-Curb	66.93	81.48	35.29	86.32	72.29	85.15	86.93	91.16	81.63
OA(%)	77.79	87.51	88.25	93.93	94.34	94.76	94.88	95.00	95.09
AA(%)	51.51	78.60	68.47	84.09	81.48	84.30	83.96	86.21	84.84
Kappa(%)	70.31	86.88	84.40	91.96	92.51	93.07	93.21	93.38	93.50

strategy is designed to calculate the total loss of two individual and cross-modal branches. The MFT improves the generalization ability of the model by utilizing a multihead cross-patch attention and additional supplementary information sources. At the same time, the concept of tokenization is introduced to learn a distinctive representation in a reduced and hierarchical feature space, thereby improving the HSI land-cover classification performance. The DFINet can extract autocorrelation and cross correlation from multisource feature pairs, thus highlighting meaningful complementary information for classification. At the same time, the consistency loss, discrimination loss, and classification loss are coordinated to improve the classification accuracy. The NNCNet uses unlabeled data through self-supervised learning to overcome the problem of inconsistent distribution of multisource data. It uses nearest neighbor data enhancement and bilinear attention modules to enhance semantic similarity and feature interaction, respectively, improving classification results. The ISSP addresses the problem of effectively utilizing complementary information from PAN and MS images by using pixel-guided spatial enhancement and time–frequency cooperative spectral enhancement. The 3DSADiff is a multimodal diffusion model that utilizes a spatial–spectral joint denoising network to enhance the quality of diffusion features. In addition, it strengthens feature representation capabilities through a 3-D self-awareness module, thereby achieving the fusion and classification of RS images.

Similar to the approaches discussed in this article, these methods all aim to improve classification performance by fusing MS and PAN image features. However, they generally fail to address the significant redundancy present in MS and PAN features and employ relatively coarse fusion techniques to combine the two modalities. Building upon this, this article effectively reduces feature redundancy and achieves efficient fusion of the two modalities through the proposed CDAM and ALFM.

1) Experiment Results and Analysis on Xi'an Urban Area: The proposed DASF-Model achieved top-three rankings in nearly all metrics on Xi'an Urban dataset, with the highest classification accuracy in c_6 (Tree: 93.79%). It attained the highest OA (85.35%), AA (79.20%), and Kappa (81.14%) while maintaining relatively few parameters (0.49M) and shorter runtime (40.36 s). In contrast, although the DFINet achieved the highest classification accuracy in several categories, its performance in other categories was too low, resulting in poor overall performance. Meanwhile, the

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NNCNet, despite not ranking in the top three for accuracy in almost any individual category, demonstrated stable overall performance, enabling it to excel in both OA (83.98%), AA (76.26%), and Kappa (79.31%). Finally, the classification result maps and misclassification maps also visually demonstrate the superior performance of the proposed DASF-Model.

2) Experiment Results and Analysis on Nanjing Area: The DASF-Model leverages exceptional feature interaction capabilities, leading to optimal classification accuracy in c_7 (Intensive Building: 77.07%). Notably, it also demonstrates superior performance with an OA of 79.11%, AA of 73.92%, and Kappa of 75.54%. The CRHFF utilizes an encoder to extract multichannel and high-resolution features, excelling in global background feature extraction and yielding impressive results in c_2 (Road1: 72.96%). The spatial location capturing ability of AM^{3}Net contributes to excellent categorization in c_8 (White Building: 90.98%), c_9 (Sparse Building: 80.99%), and c_{10} (Road2: 91.86%). It should be noted that the model exhibits significant performance differences between Road1 and Road2 in Nanjing dataset. This is because Road2 represents wetland paths along the riverbank, while Road1 corresponds to urban roads. Therefore, this discrepancy is reasonable. The DFINet exhibits spectral collection capabilities, reflecting superior performance in c_1 (Water: 96.18%), c_4 (Vegetation: 88.77%), c_5 (Red Building: 85.50%), and c_6 (Blue Building: 97.09%). The NNCNet enhances neighborhood information using a bilinear attention module, achieving peak accuracy in category c_{11} (Low vegetation: 66.78%). The ISSP effectively integrates multiscale context information while emphasizing pixel-level features and achieves the highest accuracy in category c_3 (Bareland: 94.50%).

3) Experiment Results and Analysis on Xi'an Area: By comparing the performance of different models, it can be seen that the DASF-Model performs best in overall performance indicators, with an OA of 80.56%, AA of 73.93%, and Kappa of 77.74%, all higher than other models. Specifically, for each category, the performance of each model in different categories is different. For example, the MFT excels in c_1 (Water: 97.35%), while the NNCNet demonstrates superior performance in c_2 (Road1: 34.16%). In other categories, the AM^{3}Net demonstrates the highest accuracy in c_6 (Farmland: 76.35%), c_9 (Sparse Building: 80.85%), and c_{10} (Suburb Building: 99.81%), benefiting from its involution operator. Leveraging deep cross-attention mechanisms, the DFINet attains superior performance in c_4 (Bareland: 75.55%), c_7 (Tree: 92.29), c_8 (Intensive Building: 86.75%), and c_{12} (White Building: 83.25%). Meanwhile, the CRHFF and the ISSP demonstrate superior performance in c_5 (Lowvegetation: 55.51%), c_{11} (Workshop Building: 95.39%), and c_3 (Road2: 68.75%), respectively. Moreover, with its sophisticated diffusion architecture and 3-D self-awareness module, 3DSADiff achieves the highest accuracy in c_3 (Road2: 75.26%) while attaining suboptimal performance in OA (80.33%), AA (73.89%), and Kappa (77.49%). Although the DASF-Model does not achieve the highest accuracy in any specific category, it consistently maintains competitive performance across all classes. This suggests that our approach maintains robust classification capability across all categories rather than excelling in specific classes at the expense of others. These results further prove that the overall classification performance of the DASF-Model on Xi'an dataset is better than other models and has better generalization ability and robustness.

4) Experiment Results and Analysis on MUUFL Area: On the MUUFL dataset, the proposed DASF-Model achieved the highest accuracy in c_3 (Road: 93.69%), c_5 (Sidewalk: 97.21%), and c_6 (Cloth-Panels: 98.12%), while also demonstrating strong classification performance in other categories. As a result, it attained the highest scores in both OA (95.09%) and Kappa (93.50%). In comparison, the S^{2}ENet delivered the best performance in several categories but suffered from low accuracy in others, leading to subpar overall and AA. Meanwhile, 3DSADiff employs a spatial–spectral joint denoising network to enhance diffusion feature quality, consequently achieving the highest classification accuracy in categories c_1 (Tree: 98.47%), c_{10} (Buildings: 34.96%), and c_{11} (Yellow-Curb: 91.16%). Meanwhile, it demonstrates outstanding performance across all three global evaluation metrics, particularly attaining the highest score in AA (86.21%).

Quantitative assessments reveal that the DASF-Model surpasses most reference algorithms in classification accuracy, with rigorous testing across multiple datasets confirming its state-of-the-art performance metrics.

V. Conclusion

In this article, a DASF-Model was proposed. It mainly included the CDAM and the ALFM. The CDAM used a diff-attention mechanism to effectively separate the common features and dominant features of MS and PAN images, reducing the redundancy of the feature fusion stage. At the same time, SPVM was used to capture the global and local features of the image, and the global context relationship between independent tokens was modeled. The ALFM effectively fused features with large semantic differences while keeping the size of fusion features unchanged by calculating the overall pixel-by-pixel influence of one modality on the other modality. The effectiveness and robustness of the DASF-Model were demonstrated through experiments on four datasets.

The main contribution of this work is the innovative use of SSMs to effectively separate shared features and modality-specific dominant features between MS and PAN images, thereby avoiding redundant fusion of common features during the integration phase and achieving feature deredundancy. In addition, the proposed attention-driven pixelwise linear fusion method enables effective integration of both shared and dominant features. However, this study's limitations primarily manifest in: 1) the current approach of using parameter matrix swapping for feature separation and redundancy removal cannot be directly extended to handle three or more input modalities, resulting in poor generalizability for more complex multimodal scenarios and 2) when input modalities exhibit low correlation or lack shared essential features, the proposed method may underperform due to its excessive focus on feature separation and redundancy removal.

Future research will focus on enhancing the model's generalization capabilities while maintaining its current level of

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precision, particularly for deployment in more diverse real-world scenarios.

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